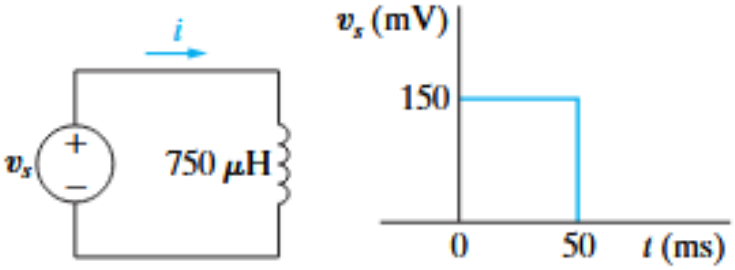
EE 2010
Fall 2026

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Game 28
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Inductor:	$v_L = L\frac{di}{dt}$	$\rightarrow$	$i_L(t) = \frac{1}{L}\int_{t_0}^{t} v\, d\tau + i(t_0)$
Capacitor:	$i_C = C\frac{dv}{dt}$	$\rightarrow$	$v_c(t) = \frac{1}{C}\int_{t_0}^{t} i\, d\tau + v(t_0)$

a) Given the simple inductor circuit below.
Find $i(t)$ for all periods of time ($t < 0, 0 < t < 50\ ms, t > 50\ ms$)



b) The voltage across a 5 μF capacitor is known to be

$$v_c = 500te^{-2500t}\ V \qquad for \quad t \geq 0$$

Find the current through the capacitor for $t > 0$. Assume the passive sign convention.

a) $t < 0$ $i(t) = 0A$

$0 < t < 50$ $\frac{1}{750\mu H}\int_0^t 0.15\, dt + 0 = \frac{0.15}{0.000750} = 200At$

$50 < t$ $\frac{1}{0.000750H}\int_{50}^{t} 0\, dt + 200(0.05)$

$= \frac{1}{0.000750}(0) + 10A = 10At$

$v_c = 500te^{-2500t}$

$i_c = C\frac{dv_c}{dt} = C(-500e^{-2500t}(2500t - 1))$

b) $= -0.0025e^{-2500t}(2500t - 1)$